ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

School of Engineering			
Department of Computer Science and Engineering			
Course Code:		Semester:	V
Course Title:	AI & Machine Learning	Year:	2024 - 2025
Faculty Name:	Prof. Sasikala N		

MODULE 1

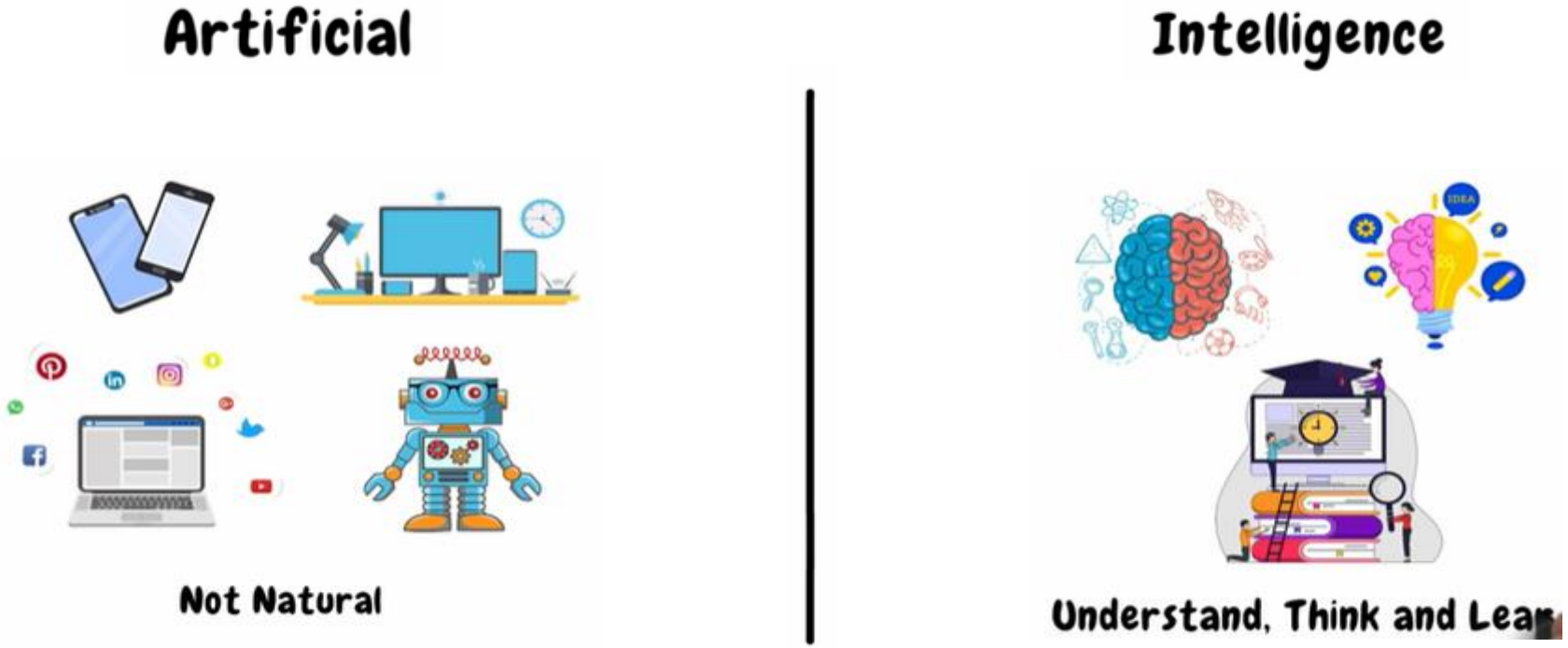
Syllabus

INTRODUCTION: What Is AI, The Foundations of Artificial Intelligence, The History of Artificial Intelligence, The State of the Art, Risks and Benefits of AI

INTELLIGENT AGENTS: Agents and Environments, Good Behavior: The Concept of Rationality, The Nature of Environments, The Structure of Agents

(Text Book 1: Chapter 1, 2)

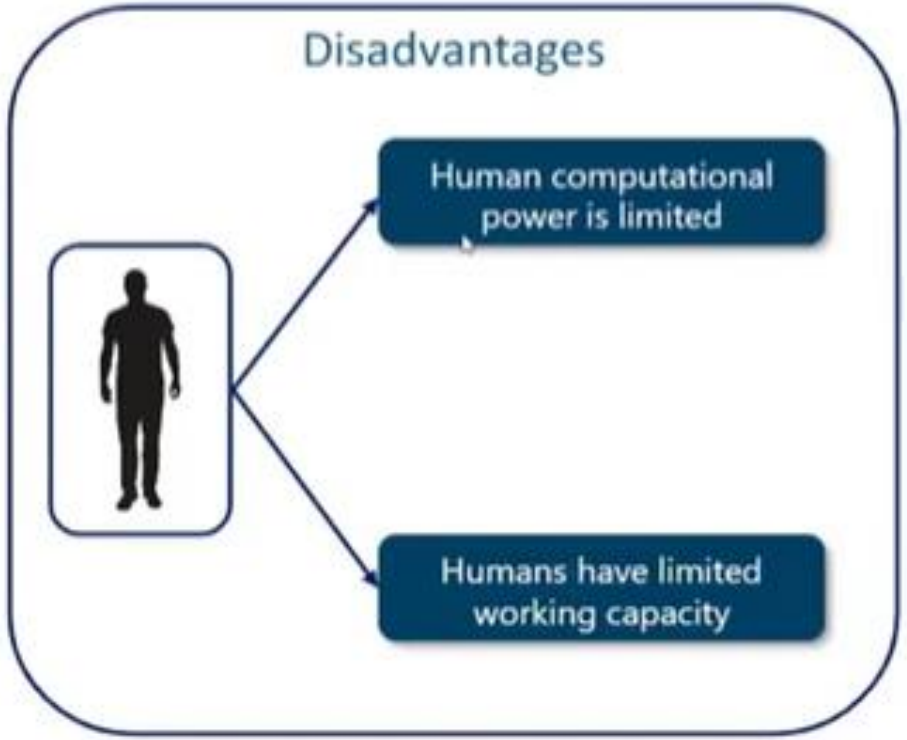
What is Artificial Intelligence?

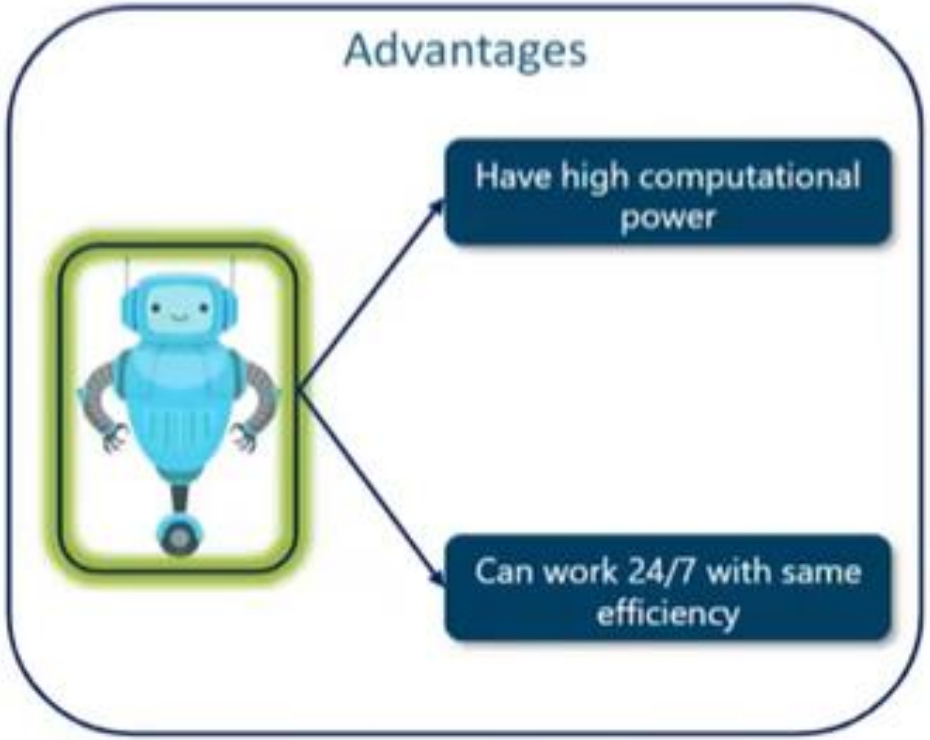


What is Artificial Intelligence?

Artificial + Intelligence?

What is Artificial Intelligence?





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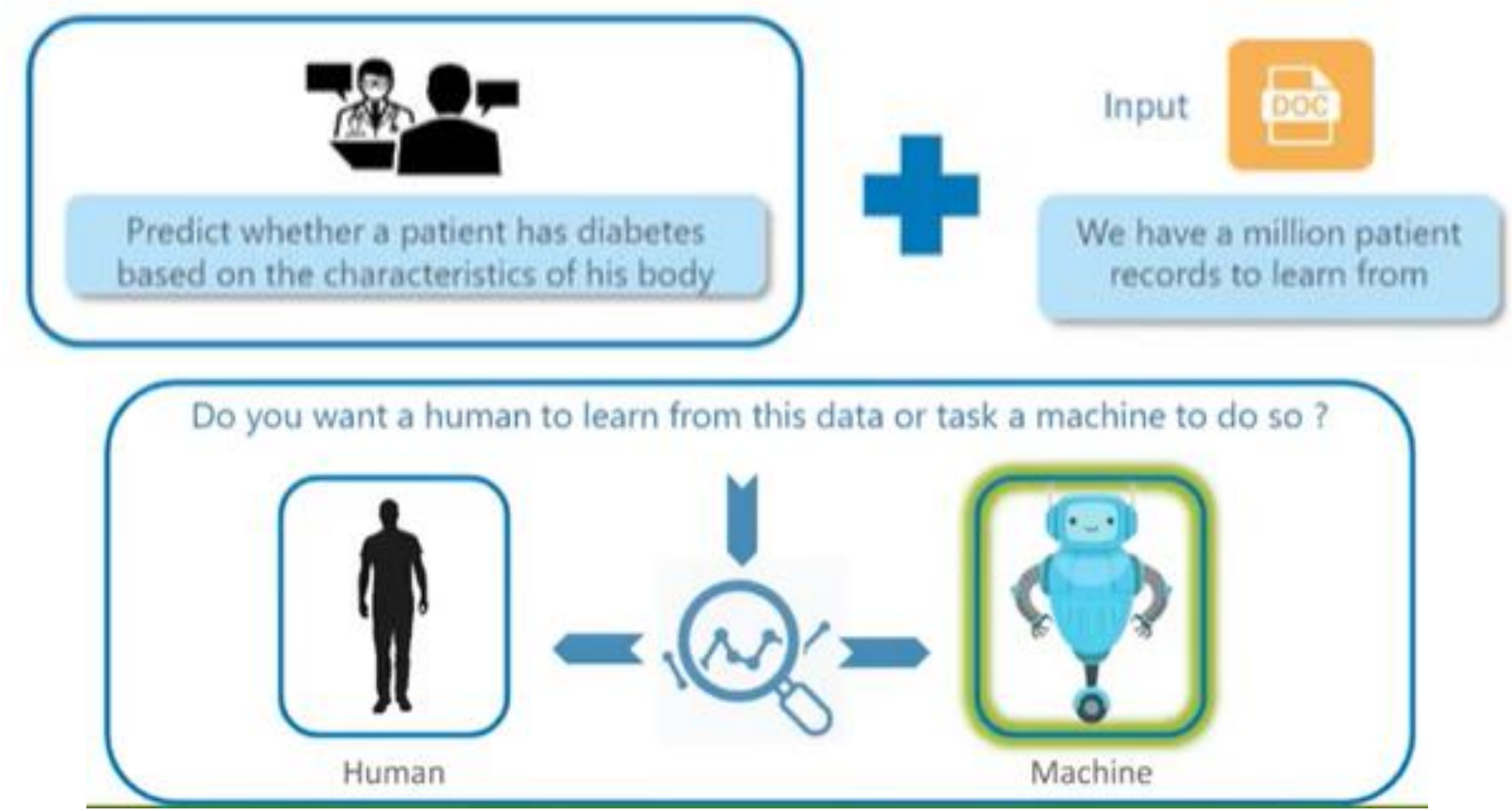


What is Artificial Intelligence?

According to the father of Artificial Intelligence, John McCarthy, it is *"The science and engineering of making intelligent machines, especially intelligent computer programs"*.

AI holds a tendency to cause a machine to work as a human.

Why Artificial Intelligence?

- With the help of AI, you can create such software or devices which can ***solve real-world problems very easily and with accuracy*** such as health issues, marketing, traffic issues, etc.
- With the help of AI, you can create your personal virtual Assistant, such as Cortana, Google Assistant, Siri, etc.
- With the help of AI, you can build such Robots which can work in an environment where survival of humans can be at risk.
- AI opens a path for other new technologies, new devices, and new Opportunities.

AI Domains

Machine playing chess

Game Playing

Programming computers to play games against human opponents

Google's Self driving car

Expert Systems

Programming computers to make decisions in real-life situations

Natural Language Processing

Programming computer systems to understand natural human languages

Handwriting recognition

Robotics

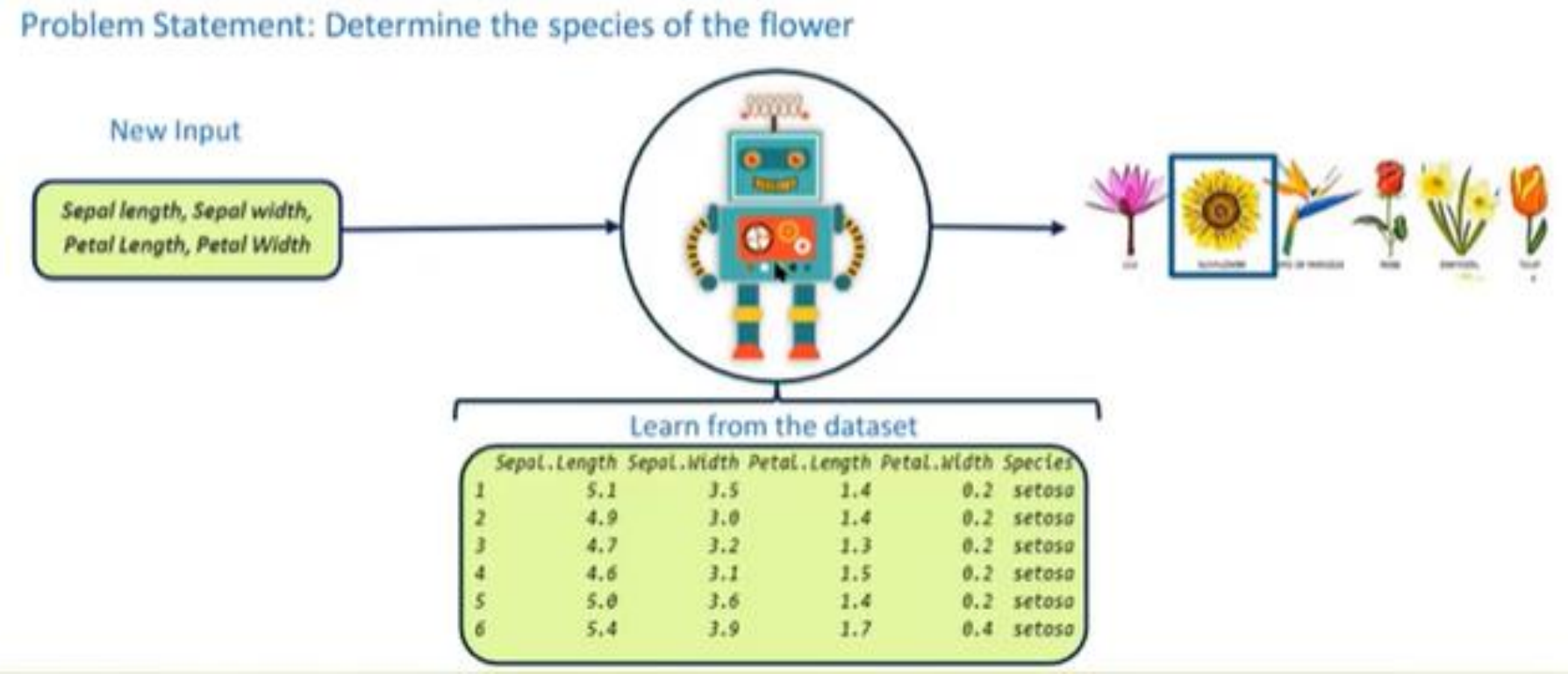
Programming computers to see and hear and react according to sensors incorporated in them

Artificial intelligence program - Siri

AI Subsets

Artificial Intelligence
Machine Learning
Deep Learning
ARTIFICIAL INTELLIGENCE
A technique which enables machines to mimic human behaviour
MACHINE LEARNING
Subset of AI technique which use statistical methods to enable machines to improve with experience.
DEEP LEARNING
Subset of ML which make the computation of multi-layer neural network feasible

- Type of AI that provide computers with the ability to learn without being explicitly programmed.



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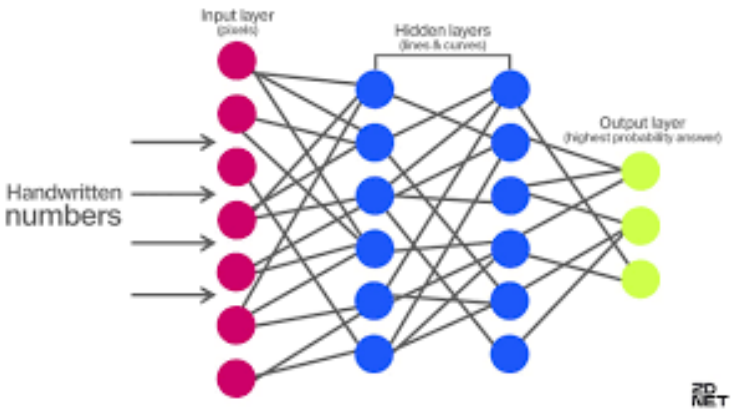
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

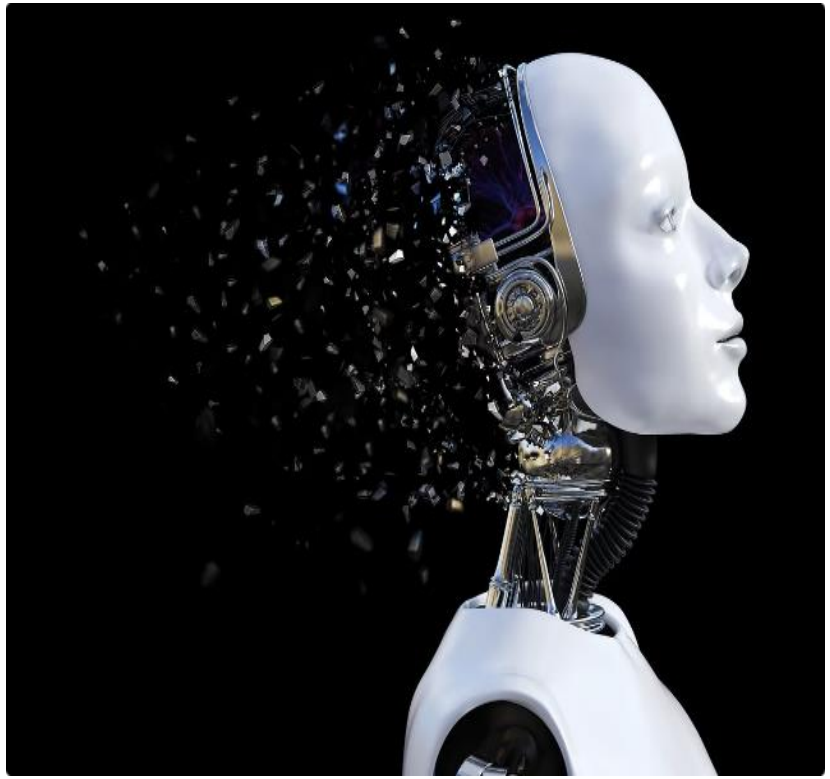
- Deep Learning

Deep learning is the subset of machine learning methods based on neural networks with representation learning.

The adjective "deep" refers to the use of multiple layers in the network.



INTRODUCTION



Intelligence - how we think and act—that is, how our brain, a mere handful of matter, can perceive, understand, predict, and manipulate a world far larger and more complicated than itself.

Artificial intelligence - concerned with not just understanding but also building intelligent entities—machines that can compute how to act effectively and safely in a wide variety of novel situations.

AI expert Kai-Fu Lee predicts that its impact will be "more than anything in the history of mankind."

Machine learning is a subfield of AI that studies the ability to improve performance based on experience.

Some AI systems use machine learning methods to achieve competence, but some do not.

human decisions are not always mathematically perfect.

What Is AI?

Some have defined intelligence in terms of fidelity to human performance, while others prefer an abstract, formal definition of intelligence called rationality—loosely speaking, doing the “right thing.”

two dimensions—human vs. rational and thought vs. behavior—there are four possible combinations

(Artificial intelligence is the simulation of human intelligence processes by machines, especially computer systems)

Four approaches

1. Acting humanly: The Turing test approach

The Turing test, proposed by Alan Turing (1950), was designed as a thought experiment that would sidestep the philosophical vagueness of the question “Can a machine think?”

The computer would need the following capabilities:

- natural language processing to communicate successfully in a human language;
- knowledge representation to store what it knows or hears;
- automated reasoning to answer questions and to draw new conclusions;
- machine learning to adapt to new circumstances and to detect and extrapolate patterns.

cont..

A total Turing test - requires interaction with objects and people in the real world.

To pass the total Turing test, a robot will need

computer vision and speech recognition to perceive the world

robotics to manipulate objects and move about

2. Thinking humanly: The cognitive modeling approach

A program thinks like a human, we must know how humans think.

We can learn about human thought in three ways:

introspection—trying to catch our own thoughts as they go by;

psychological experiments—observing a person in action;

brain imaging—observing the brain in action.

The interdisciplinary field of cognitive science brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.

3. Thinking rationally: The “laws of thought” approach

These laws of thought were supposed to govern the operation of the mind; their study initiated the field called logic.

Logicist tradition within artificial intelligence hopes to build on such programs to create intelligent systems.

The theory of probability fills this gap, allowing rigorous reasoning with uncertain information.

4. Acting rationally: The rational agent approach

An agent is just something that acts (agent comes from the Latin agere, to do).

Of course, all computer programs do something, but computer agents are expected to do more: operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals.

A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

The rational-agent approach to AI has two advantages over the other approaches.

First, it is more general than the “laws of thought” approach because correct inference is just one of several possible mechanisms for achieving rationality.

Second, it is more amenable to scientific development.

In a nutshell, AI has focused on the study and construction of agents that do the right thing.

What counts as the right thing is defined by the objective that we provide to the agent. This general paradigm is so pervasive that we might call it the standard model.

Beneficial machines

- The standard model has been a useful guide for AI research since its inception, but it is probably not the right model in the long run.
- The reason is that the standard model assumes that we will supply a fully specified objective to the machine.
- For an artificially defined task such as chess or shortest-path computation, the task comes with an objective built in—so the standard model is applicable.
- As we move into the real world, however, it becomes more and more difficult to specify the objective completely and

Beneficial machines

For example, in designing a self-driving car, one might think that the objective is to reach the destination safely.

But driving along any road incurs a risk of injury due to other errant drivers, equipment failure, and so on; thus, a strict goal of safety requires staying in the garage.

There is a tradeoff between making progress towards the destination and incurring a risk of injury.

How should this tradeoff be made? Furthermore, to what extent can we allow the car to take actions that would annoy other drivers? How much should the car moderate its acceleration, steering, and braking to avoid shaking up the passenger?

These kinds of questions are difficult to answer a priori. They are particularly problematic in the general area of human–robot interaction, of which the self-driving car is one example.

The problem of achieving agreement between our true preferences and the objective we put into the machine is called the value alignment problem: the values or objectives put into the machine must be aligned with those of the human.

https://www.moralmachine.net/

When a machine knows that it doesn't know the complete objective, it has an incentive to act cautiously, to ask permission, to learn more about our preferences through observation, and to defer to human control. Ultimately, we want agents that are provably beneficial to humans.

The Foundations of Artificial Intelligence

1. Philosophy

Can formal rules be used to draw valid conclusions?

How does the mind arise from a physical brain?

Where does knowledge come from?

How does knowledge lead to action?

Dualism - a part of the human mind (or soul or spirit) that is outside of nature, exempt from physical laws.

2. Mathematics

What are the formal rules to draw valid conclusions?

What can be computed?

How do we reason with uncertain information?

The idea of formal logic can be traced back to the philosophers of ancient Greece, India, and China, but its mathematical development really began with the work of George Boole, who worked out the details of propositional, or Boolean, logic.

Gottlob Frege extended Boole's logic to include objects and relations, creating the first-order logic that is used today.

The theory of probability can be seen as generalizing logic to situations with uncertain information—a consideration of great importance for AI.

The formalization of probability, combined with the availability of data, led to the emergence of statistics as a field.

The history of computation is as old as the history of numbers, but the first nontrivial algorithm is thought to be Euclid's algorithm for computing greatest common divisors.

Incompleteness theorem showed that in any formal theory as strong as Peano arithmetic (the elementary theory of natural numbers), there are necessarily true statements that have no proof within the theory.

Alan Turing - characterize exactly which functions are computable - capable of being computed by an effective procedure.

Although computability is important to an understanding of computation, the notion of tractability has had an even greater impact on AI.

The theory of NP-completeness, provides a basis for analyzing the tractability of problems: any problem class to which the class of NP complete problems can be reduced is likely to be intractable.

3. Economics

How should we make decisions in accordance with our preferences?

How should we do this when others may not go along?

How should we do this when the payoff may be far in the future?

The science of economics - An Inquiry into the Nature and Causes of the Wealth of Nations.

Decision theory, which combines probability theory with utility theory, provides a for mal and complete framework for individual decisions (economic or otherwise) made under uncertainty—that is, in cases where probabilistic descriptions appropriately capture the decision maker's environment.

Game theory included the surprising result that, for some games, a rational agent should adopt policies that are randomized.

4. Neuroscience

How do brains process information?

Neuroscience is the study of the nervous system, particularly the brain.

- the brain consisted largely of nerve cells or neurons
- advances in single cell electrical recording of neuron activity and by the methods of optogenetics, which allow both measurement and control of individual neurons modified to be light-sensitive.
- The development of brain–machine interfaces for both sensing and motor control not only promises to restore function to disabled individuals, but also sheds light on many aspects of neural systems.
- Futurists make much of these numbers, pointing to an approaching singularity at which computers reach a super human level of performance and then rapidly improve themselves even further.

5. Psychology

How do humans and animals think and act?

behaviorism movement, rejected any theory involving mental processes on the grounds that introspection could not provide reliable evidence.

Cognitive psychology, which views the brain as an information-processing device.

three key steps of a knowledge-based agent:

(1) the stimulus must be translated into an internal representation,

(2) the representation is manipulated by cognitive processes to derive new internal representations,

(3) these are in turn retranslated back into action.

6. Computer engineering

How can we build an efficient computer?

- Each generation of computer hardware has brought an increase in speed and capacity and a decrease in price—a trend captured in Moore's law

- quantum computing holds out the promise of far greater accelerations for some important Quantum computing subclasses of AI algorithms.

7. Control theory and cybernetics

How can artifacts operate under their own control?

Control System, Feedback loops, Communication, Self Organization

control theory - biological and mechanical control systems and their connection to cognition.

Use of homeostatic devices containing appropriate feedback loops to achieve stable adaptive behavior.

Modern control theory, especially the branch known as stochastic optimal control, has as its goal the design of systems that maximize a cost function over time.

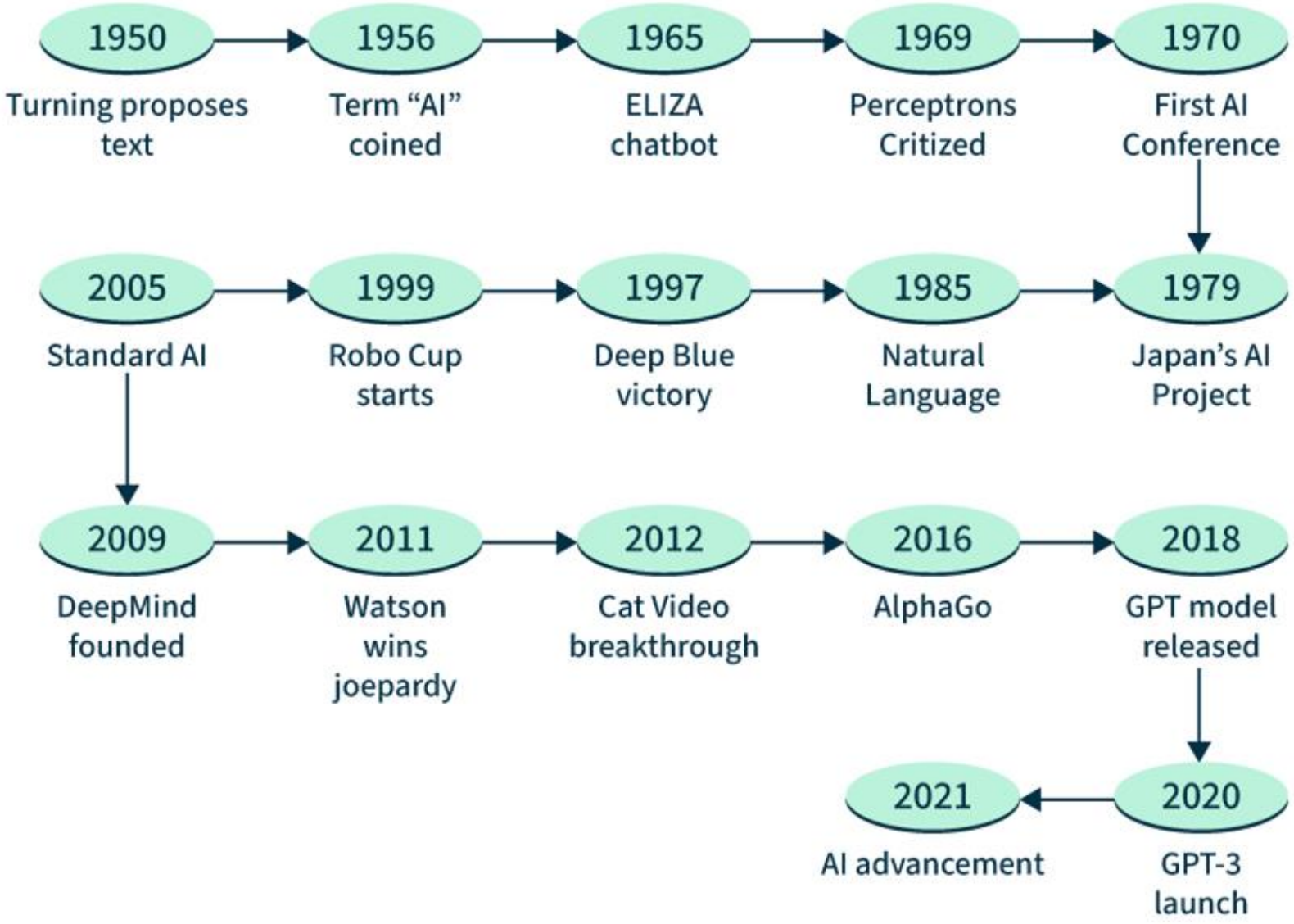
8. Linguistics

How does language relate to thought?

Modern linguistics and AI, then, were “born” at about the same time, and grew up together, intersecting in a hybrid field called computational linguistics or natural language.

knowledge representation (the study of how to put knowledge into a form that a computer can reason with) was tied to language and informed by research in linguistics, which was connected in turn to decades of work on the philosophical analysis of language.

The History of Artificial Intelligence



History of Artificial Intelligence (1940-1950)

- **1943**: Warren McCulloch and Walter Pitts develop a mathematical model of artificial neurons, laying the foundation for neural networks.
- **1949**: Donald Hebb proposes a learning theory for neurons, later known as Hebbian learning, influencing neural network development.
- **1950**: Alan Turing introduces the Turing Test, a criterion for machine intelligence.
- **1957**: Frank Rosenblatt develops the Perceptron, an early model for supervised learning in neural networks.
- **1958**: John McCarthy invents the Lisp programming language, which becomes crucial for AI research.

History of Artificial Intelligence (1960)

- **1961**: The first industrial robot, Unimate, is introduced, revolutionizing manufacturing.
- **1963**: MIT receives a $2.2 million grant from DARPA to research AI, marking significant government investment in the field.
- **1965**: Joseph Weizenbaum creates ELIZA, an early natural language processing program simulating a psychotherapist.

History of Artificial Intelligence (1970)

- **1971**: The first expert system, DENDRAL, is developed for chemical analysis, demonstrating AI's potential in specialized fields.
- **1972**: The programming language Prolog is developed, becoming central to AI research in logic programming and expert systems.
- **1973**: The "Lighthill Report" criticizes AI research, leading to reduced funding in the UK and contributing to the first AI Winter.
- **1979**: The Stanford Cart, an early autonomous vehicle, successfully navigates a room full of chairs without human intervention.

The History of Artificial Intelligence (1980)

- **1980**: The Japanese government announces the Fifth Generation Computer Systems project, aiming to create intelligent computers.
- **1982**: John Hopfield popularizes the Hopfield network, a form of recurrent neural network for associative memory.
- **1986**: The backpropagation algorithm is popularized, leading to a resurgence in neural network research.
- **1987**: The market for Lisp machines, specialized AI hardware, collapses, contributing to the second AI Winter.

History of Artificial Intelligence (1990)

- **1990**: Rodney Brooks introduces behavior-based robotics, emphasizing decentralized control over traditional AI approaches.
- **1994**: The first conference on Knowledge Discovery and Data Mining (KDD) is held, highlighting the importance of AI in data analysis.
- **1996**: IBM's Deep Blue defeats world chess champion Garry Kasparov in one of their matches, showcasing AI's prowess in strategic games.
- **1998**: The AI language model LDA (Latent Dirichlet Allocation) is introduced, advancing natural language processing and topic modeling.

History of Artificial Intelligence (2000)

- **2000**: Honda's ASIMO, a humanoid robot, is introduced, demonstrating advanced robotics and AI in human-robot interaction.
- **2001**: Aibo, Sony's robotic dog, gains popularity, showing the potential for AI in consumer entertainment and companionship.
- **2004**: DARPA's Grand Challenge, a competition for autonomous vehicles, pushes advancements in self-driving car technology.
- **2009**: Google develops the first self-driving car, demonstrating the practical application of AI in transportation.

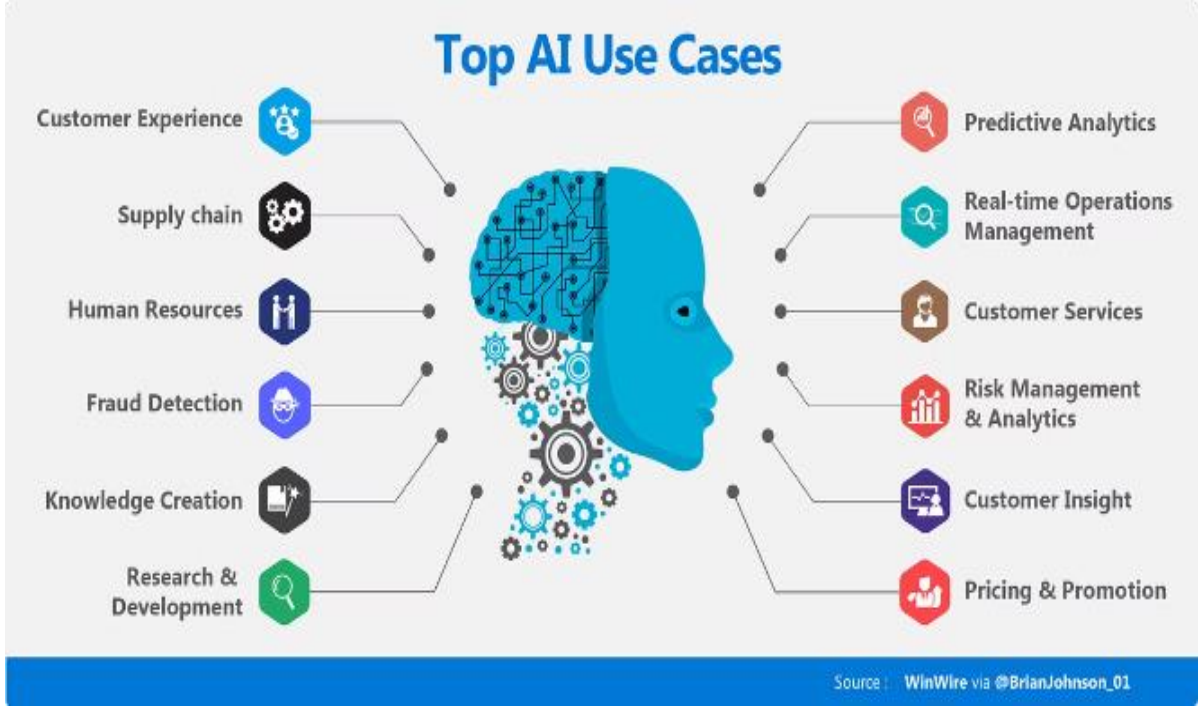
History of Artificial Intelligence (2010)

- **2010**: Microsoft launches Kinect, an AI-powered motion-sensing device, popularizing gesture recognition in gaming.
- **2012**: AlexNet wins the ImageNet competition, significantly improving image recognition accuracy and popularizing deep learning.
- **2015**: OpenAI is founded to advance digital intelligence safely and broadly.
- **2016**: DeepMind's AlphaGo defeats Go champion Lee Sedol, demonstrating AI's ability to master complex games with deep learning.
- **2017**: The Transformer architecture is introduced, revolutionizing natural language processing and leading to models like GPT and BERT.
- **2018**: Google Duplex, an AI that can carry out phone conversations to book appointments, is demonstrated, blurring lines between human and machine interaction.

History of Artificial Intelligence (2020)

- **2020**: GPT-3, a state-of-the-art language model, is released by OpenAI, demonstrating unprecedented language generation capabilities.
- **2021**: DeepMind's AlphaFold achieves significant breakthroughs in protein folding, revolutionizing biological research.
- **2022**: DALL-E 2, an advanced image generation model by OpenAI, is released, showcasing the potential of AI in creative applications.
- **2023**: AI technologies, including generative models, become increasingly integrated into everyday applications, from art creation to autonomous systems.

The State of the Art



1 Machine Learning Advancements

Recent breakthroughs in deep learning and other machine learning techniques have enabled AI systems to tackle increasingly complex problems, from image recognition to natural language processing and decision-making.

2 Autonomous Systems

AI-powered autonomous systems, such as self-driving cars and drones, are becoming more advanced and reliable, with the potential to revolutionize transportation, logistics, and various other industries.

The State of the Art of AI

3 AI Assistants and Chatbots

Virtual assistants like Siri, Alexa, and Google Assistant have become ubiquitous, providing users with a wide range of services, from information lookup to task automation and personalized recommendations.

4 Predictive Analytics and Decision-Making

AI algorithms are being deployed in fields like healthcare, finance, and marketing to analyze large datasets, identify patterns, and make predictions that can inform decision-making and optimize processes.

- Robotic vehicles: The history of robotic vehicles stretches back to radio-controlled cars of the 1920s, but the first demonstrations of autonomous road driving without special guides occurred in the 1980s
- Legged locomotion: BigDog, a quadruped robot by Raibert et al. (2008), upended our notions of how robots move—no longer the slow, stiff-legged, side-to-side gait of Hollywood movie robots, but something closely resembling an animal and able to recover when shoved or when slipping on an icy puddle.
- Autonomous planning and scheduling: A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft.

- Machine translation: Online machine translation systems now enable the reading of documents in over 100 languages, including the native languages of over 99% of humans, and render hundreds of billions of words per day for hundreds of millions of users.
- Speech recognition: In 2017, Microsoft showed that its Conversational Speech Recognition System had reached a word error rate of 5.1%, matching human performance on the Switchboard task, which involves transcribing telephone conversations.
- Recommendations: Companies such as Amazon, Facebook, Netflix, Spotify, YouTube, Walmart, and others use machine learning to recommend what you might like based on your past experiences and those of others like you.

- Game playing: When Deep Blue defeated world chess champion Garry Kasparov in 1997, defenders of human supremacy placed their hopes on Go.
- Piet Hut, an astrophysicist and Go enthusiast, predicted that it would take “a hundred years before a computer beats humans at Go—maybe even longer.”
- But just 20 years later, ALPHAGO surpassed all human players (Silver et al., 2017). Ke Jie, the world champion, said, “Last year, it was still quite human-like when it played.

- Image understanding: Not content with exceeding human accuracy on the challenging ImageNet object recognition task, computer vision researchers have taken on the more difficult problem of image captioning.
- Some impressive examples include “A person riding a motorcycle on a dirt road,” “Two pizzas sitting on top of a stove top oven,” and “A group of young people playing a game of frisbee”.
- Current systems are far from perfect, however: a “refrigerator filled with lots of food and drinks” turns out to be a no-parking sign partially obscured by lots of small stickers.

- Medicine: AI algorithms now equal or exceed expert doctors at diagnosing many conditions, particularly when the diagnosis is based on images.
- The widespread adoption of these techniques is now limited not by diagnostic accuracy but by the need to demonstrate improvement in clinical outcomes and to ensure transparency, lack of bias, and data privacy.
- Climate science: A team of scientists won the 2018 Gordon Bell Prize for a deep learning model that discovers detailed information about extreme weather events that were previously buried in climate data.

Risks of AI

- **Job Displacement:**
  - Automation powered by AI could lead to job losses, especially in industries relying on routine manual labor.
  - While AI creates new job opportunities, there's a risk of a skills gap as workers may not have the necessary training.
- **Bias and Discrimination:**
  - AI systems can inherit biases present in the data they are trained on, leading to unfair or discriminatory outcomes.
  - This is particularly concerning in areas like hiring, law enforcement, and credit scoring.
- **Privacy Concerns:**
  - AI can analyze vast amounts of personal data, raising concerns about privacy and surveillance.
  - The misuse of AI for data mining and profiling could lead to breaches of personal privacy.

Risks of AI

- **Security Threats:**
  - AI systems can be vulnerable to hacking, leading to potential misuse in areas like autonomous weapons or surveillance.
  - The development of AI-driven cyberattacks poses a new type of security threat.
- **Ethical Dilemmas:**
  - The use of AI in decision-making raises ethical questions, such as in autonomous vehicles deciding how to act in an accident.
  - The development of AI systems without transparency can lead to a lack of accountability.
- **Dependence on AI:**
  - Over-reliance on AI could reduce human skills and decision-making abilities.
  - There’s a risk that critical sectors like healthcare and finance could become overly dependent on AI systems, leading to vulnerabilities if those systems fail.

Benefits of AI

➢ **Efficiency and Productivity:**

- AI can automate routine tasks, improving efficiency and productivity across various industries.
- It can process large amounts of data faster than humans, leading to quicker decision- making.

➢ **Medical Advancements:**

- AI is used in diagnosing diseases, predicting patient outcomes, and personalizing treatment plans.
- AI-driven tools like image analysis help detect diseases like cancer earlier and more accurately.

➢ **Enhanced Customer Experience:**

- AI-powered chatbots and virtual assistants provide 24/7 customer service, improving user experience.
- Personalization algorithms help tailor products, services, and content to individual preferences.

Benefits of AI

- **Improved Safety:**

• AI technologies like autonomous vehicles aim to reduce human error, potentially decreasing accidents.

• AI-driven monitoring systems can predict and prevent equipment failures in industrial settings.

- **Data Analysis and Insights:**

• AI can analyze vast datasets to uncover patterns and insights that humans might miss.

• This capability is particularly valuable in fields like finance, marketing, and scientific research.

- **Innovation in New Fields:**

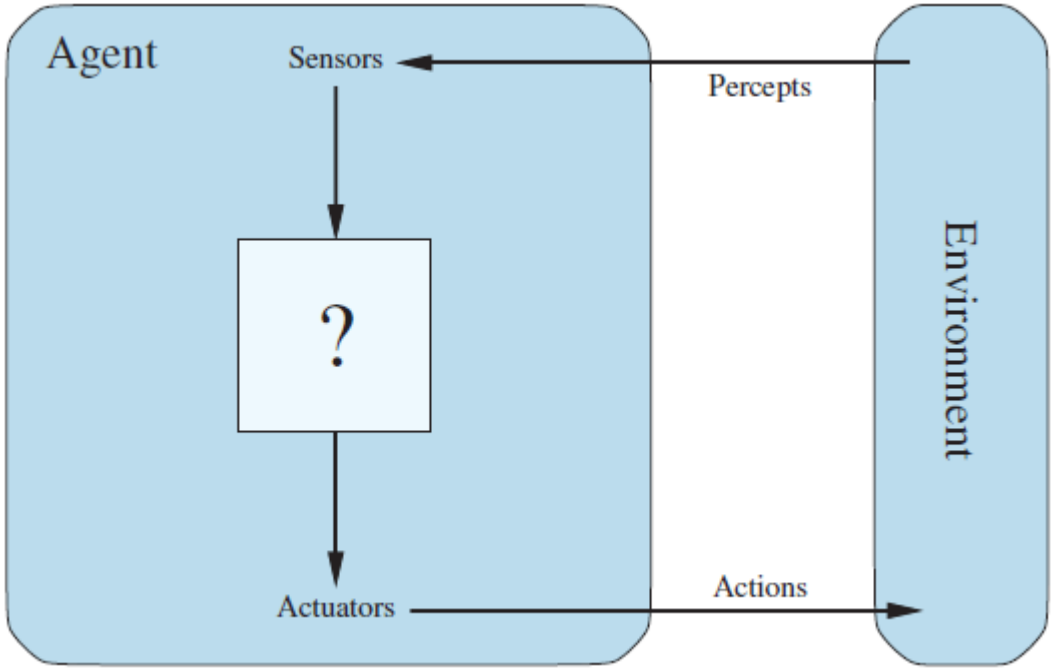
• AI is driving innovation in fields like robotics, biotechnology, and environmental science.

• It enables the development of new tools and technologies, such as smart cities and renewable energy solutions.

INTELLIGENT AGENTS

Agents and Environments

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.
- Percept to refer to the content an agent's sensors are perceiving. An agent's percept sequence is the complete history of everything the agent has ever perceived.
- Agent's behavior is described by the agent function that maps any given percept sequence to an action.



Agents interact with environments through sensors and actuators.

Imagine tabulating the agent function that describes any given agent; for most agents, this would be a very large table—infinite, in fact, unless we place a bound on the length of percept sequences we want to consider.

Given an agent to experiment with, we can, in principle, construct this table by trying out all possible percept sequences and recording which actions the agent does in response.

The table is, of course, an external characterization of the agent.

Internally, the agent function for an artificial agent will be implemented by an agent program.

It is important to keep these two ideas distinct.

The agent function is an

- abstract mathematical description;
- the agent program is a concrete implementation, running within some physical system.

Simple example—the vacuum-cleaner world, Which consists of a robotic vacuum-cleaning agent in a world consisting of squares that can be either dirty or clean.

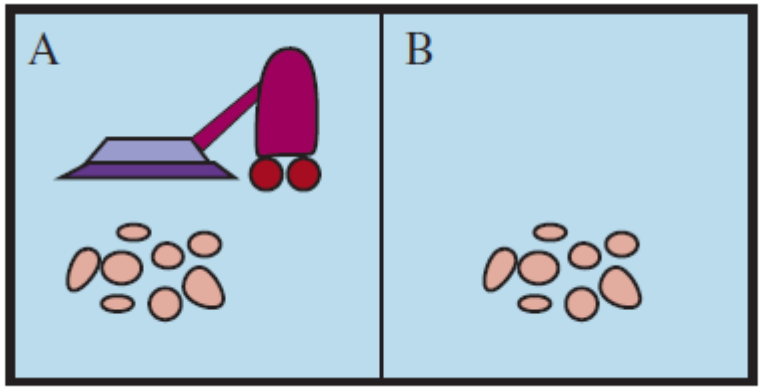


Figure shows a configuration with just two squares, A and B. The vacuum agent perceives which square it is in and whether there is dirt in the square. The agent starts in square A. The available actions are to move to the right, move to the left, suck up the dirt, or do nothing.

One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square.

agent function for the vacuum-cleaner world

Percept sequence	Action
$[A, Clean]$	*Right*
$[A, Dirty]$	*Suck*
$[B, Clean]$	*Left*
$[B, Dirty]$	*Suck*
$[A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Dirty]$	*Suck*
$\vdots$	$\vdots$
$[A, Clean], [A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Clean], [A, Dirty]$	*Suck*
$\vdots$	$\vdots$

Good Behavior: The Concept of Rationality

- A rational agent is one that does the right thing. Obviously, doing the right thing is better than doing the wrong thing, but what does it mean to do the right thing?

Performance measures

- Moral philosophy has developed several different notions of the "right thing," but AI has generally stuck to one notion called consequentialism: we evaluate an agent's behavior by its consequences. This notion of desirability is captured by a performance measure that evaluates any given sequence of environment states.
- As a general rule, it is better to design performance measures according to what one actually wants to be achieved in the environment, rather than according to how one thinks the agent should behave.

Rationality

What is rational at any given time depends on four things:

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

definition of a rational agent

- For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Omniscience, learning, and autonomy

- An omniscient agent knows the actual outcome of its actions and can act accordingly; but omniscience is impossible in reality.
- Rationality maximizes expected performance, while perfection maximizes actual performance.
- Information gathering is provided by the exploration that must be undertaken by a vacuum-cleaning agent in an initially unknown environment.

- Rational agent not only to gather information but also to learn as Learning much as possible from what it perceives. The agent's initial configuration could reflect some prior knowledge of the environment, but as the agent gains experience this may be modified and augmented. There are extreme cases in which the environment is completely known a priori and completely predictable. In such cases, the agent need not perceive or learn; it simply acts correctly.
- Rational agent should be autonomous—it should learn what it can to compensate for partial or incorrect prior knowledge.

The Nature of Environments

- The nature of the task environment directly affects the appropriate design for the agent program.

Specifying the task environment

PEAS (Performance, Environment, Actuators, Sensors)

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

PEAS description of the task environment for an automated taxi driver.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

- Virtual task environments can be just as complex as the “real” world:

Example:

- A software agent (or software robot or softbot) that trades on auction and reselling Web sites deals with millions of other users and billions of objects, many with real images.

Properties of task environments

Fully observable vs. partially observable:

- A task environment is effectively fully observable if the sensors detect all aspects that are relevant to the choice of action; relevance, in turn, depends on the performance measure.
- An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data.
- If the agent has no sensors at all then the environment is unobservable.

Single-agent vs. multiagent:

- For example, an agent solving a crossword puzzle by itself is clearly in a single-agent environment.
- For example, in chess, the opponent entity B is trying to maximize its performance measure, which, by the rules of chess, minimizes agent A's performance measure.
- Thus, chess is a competitive multiagent environment. On the other hand, in the taxi-driving environment, avoiding collisions maximizes the performance measure of all agents, so it is a partially cooperative multiagent environment. It is also partially competitive because, for example, only one car can occupy a parking space.

Deterministic vs. nondeterministic:

- If the next state of the environment is completely determined by the current state and the action executed by the agent(s), then environment is deterministic; otherwise, it is nondeterministic.
- In principle, an agent need not worry about uncertainty in a fully observable, deterministic environment.
- If the environment is partially observable, however, then it could appear to be nondeterministic.

Episodic vs. sequential:

- In an episodic task environment, the agent's experience is divided into atomic episodes. In each episode the agent receives a percept and then performs a single action. Crucially, the next episode does not depend on the actions taken in previous episodes.
- In sequential environments, the current decision could affect all future decisions.
- Chess and taxi driving are sequential: in both cases, short-term actions can have long-term consequences.

Static vs. dynamic:

- If the environment can change while an agent is deliberating, then the environment is dynamic for that agent; otherwise, it is static.
- If the environment itself does not change with the passage of time but the agent's performance score does, then we say the environment is semidynamic.

✓ Taxi driving is clearly dynamic: the other cars and the taxi itself keep moving while the driving algorithm dithers about what to do next.

✓ Chess, when played with a clock, is semidynamic.

✓ Crossword puzzles are static.

Discrete vs. continuous:

The discrete/continuous distinction applies to the state of the environment, to the way time is handled, and to the percepts and actions of the agent.

- ✓ Chess environment has a finite number of distinct states (excluding the clock). Chess also has a discrete set of percepts and actions.
- ✓ Taxi-driving actions are also continuous (steering angles, etc.). Input from digital cameras is discrete, strictly speaking, but is typically treated as representing continuously varying intensities and locations.

Known vs. unknown:

- In a known environment, the outcomes for all actions are given.
- If the environment is unknown, the agent will have to learn how it works in order to make good decisions.
- Known environment to be partially observable—for example, in solitaire card games, i know the rules but am still unable to see the cards that have not yet been turned over.
- An unknown environment can be fully observable—in a new video game, the screen may show the entire game state but i still don't know what the buttons do until I try them.

- The hardest case is partially observable, multiagent, nondeterministic, sequential, dynamic, continuous, and unknown.
- Taxi driving is hard in all these senses, except that the driver's environment is mostly known.
- Driving a rented car in a new country with unfamiliar geography, different traffic laws, and nervous passengers is a lot more exciting.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Examples of task environments and their characteristics.

The Structure of Agents

- The job of AI is to design an agent program that implements the agent function— the mapping from percepts to actions.
- Program will run on some sort of computing device with physical sensors and actuators—agent architecture.

agent = architecture + program.

The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.

```
function TABLE-DRIVEN-AGENT(percept) returns an action

    persistent: percepts, a sequence, initially empty table, a table of actions, indexed by percept sequences, initially fully
    specified

    append percept to the end of percepts

    action LOOKUP(percepts, table)

    return action
```

Agent programs

The agent program has no choice but to take just the current percept as input because nothing more is available from the environment; if the agent's actions need to depend on the entire percept sequence, the agent will have to remember the percepts.

The agent program for a simple reflex agent in the two-location vacuum environment.

```
function REFLEX-VACUUM-AGENT( [location,status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

Four basic kinds of agent programs

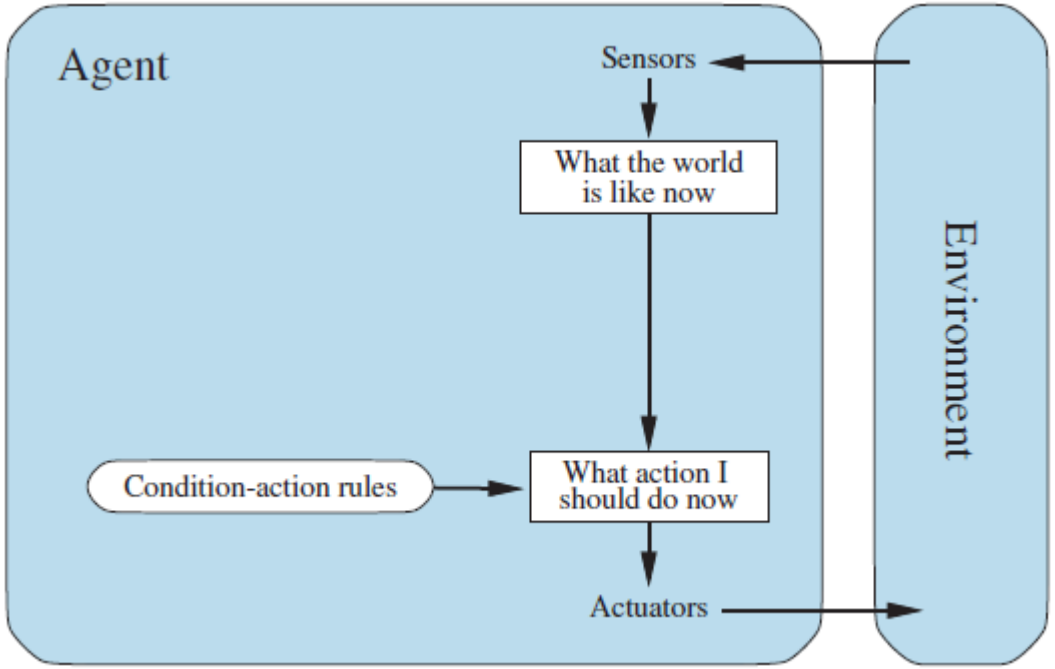
- Simple reflex agents;
- Model-based reflex agents;
- Goal-based agents;
- Utility-based agents.

Each kind of agent program combines particular components in particular ways to generate actions.

Simple reflex agents

These agents select actions on the basis Simple reflex agent of the current percept, ignoring the rest of the percept history.

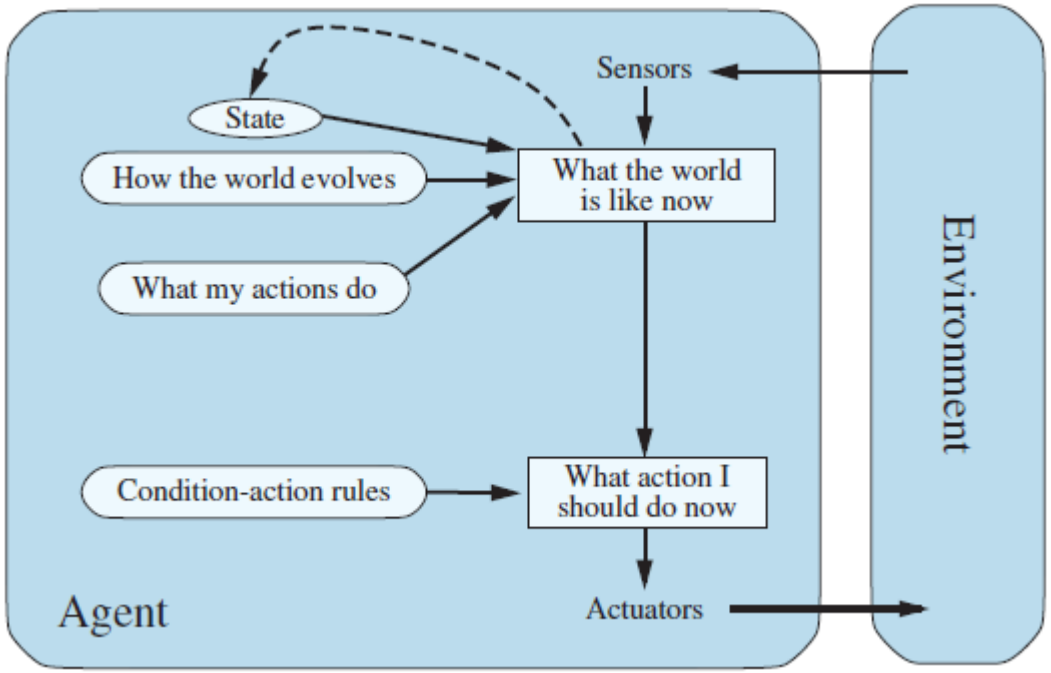
- example, the vacuum agent - its decision is based only on the current location and on whether that location contains dirt.



rectangles to denote the current internal state of the agent's decision process, and ovals to represent the background information used in the process.

Model-based reflex agents

The agent should maintain some sort of internal state that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state.



1. The effects of the agent's actions and how the world evolves independently of the agent.

- For example, when the agent turns the steering wheel clockwise, the car turns to the right, and when it's raining the car's cameras can get wet. This knowledge about "how the world works"—whether implemented in simple Boolean circuits or in complete scientific theories—is called a transition model of the world.

2. How the state of the world is reflected in the agent's percepts.

- For example, when the car in front initiates braking, one or more illuminated red regions appear in the forward-facing camera image, and, when the camera gets wet, droplet-shaped objects appear in the image partially obscuring the road. This kind of knowledge is called a sensor model.

The transition model and sensor model allow an agent to keep track of the state of the world—to the extent possible given the limitations of the agent's sensors. An agent that uses such models is called a model-based agent.

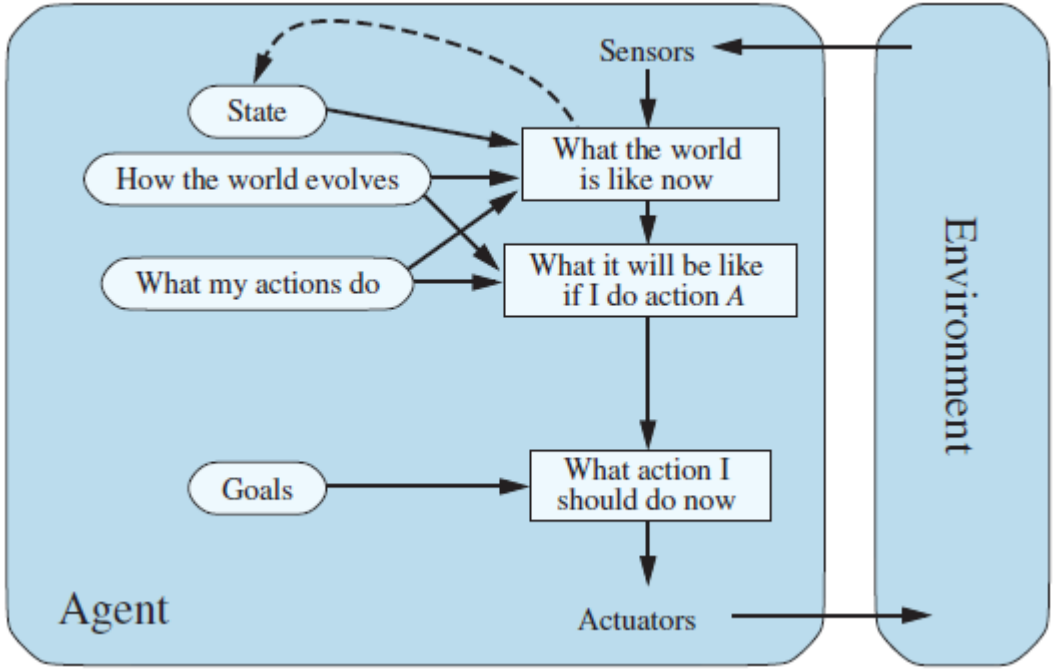
A model-based reflex agent - keeps track of the current state of the world, using an internal model.

```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
    persistent: state, the agent's current conception of the world state
                transition model, a description of how the next state depends on
                    the current state and action
                sensor model, a description of how the current world state is reflected
                    in the agent's percepts
                rules, a set of condition–action rules
                action, the most recent action, initially none
    state UPDATE-STATE(state, action, percept, transition model, sensor model)
    rule RULE-MATCH(state, rules)
    action rule. ACTION
    return action
```

Goal-based agents

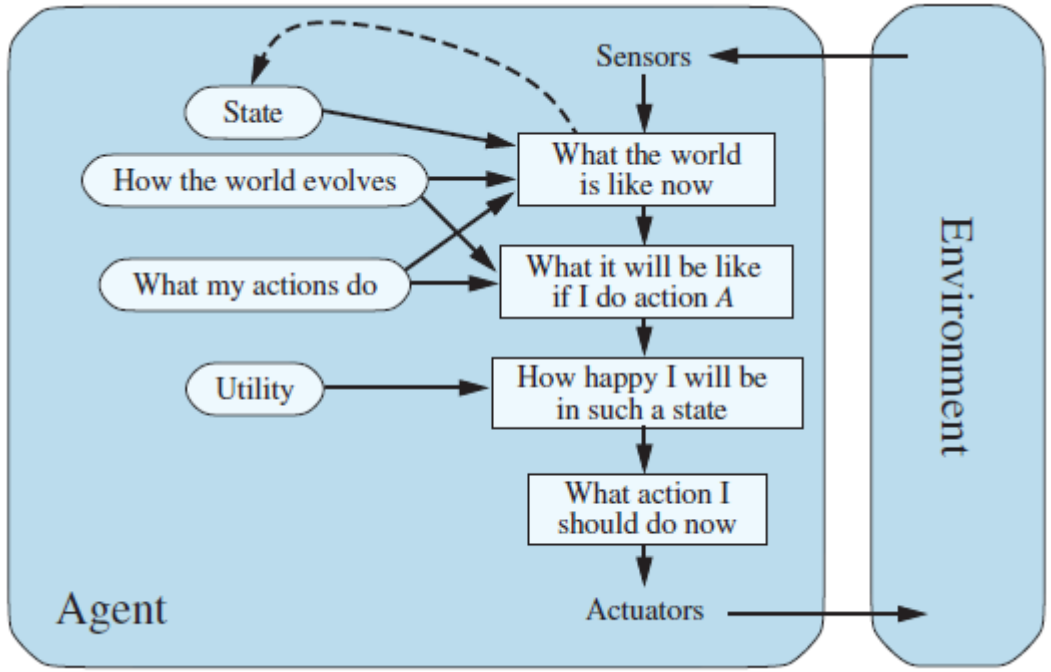
Current state description, the agent needs some sort of goal information that describes situations that are desirable.

- For example, being at a particular destination. The agent program can combine this with the model (the same information as was used in the model-based reflex agent) to choose actions that achieve the goal. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.



Utility-based agents

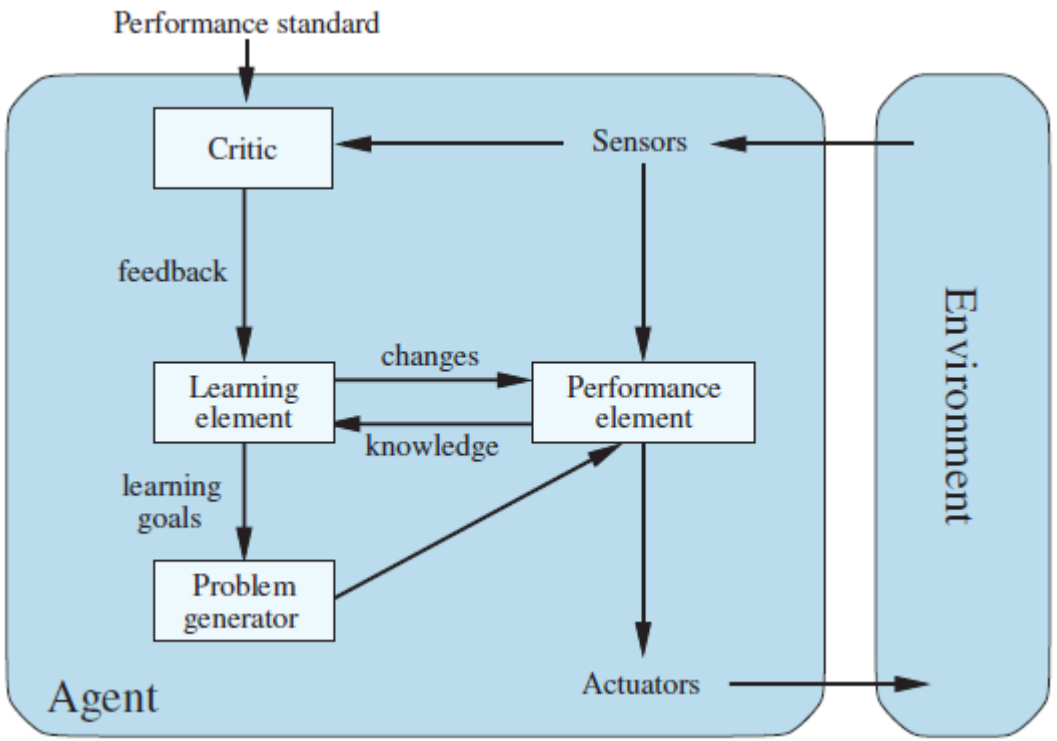
- An agent's utility function is essentially an internalization of the performance measure.
- Provided that the internal utility function and the external performance measure are in agreement, an agent that chooses actions to maximize its utility will be rational according to the external performance measure.



A rational utility-based agent chooses the action that maximizes the expected utility of the action outcomes—that is, the utility the agent expects to derive, on average, given the probabilities and utilities of each outcome.

Learning agents

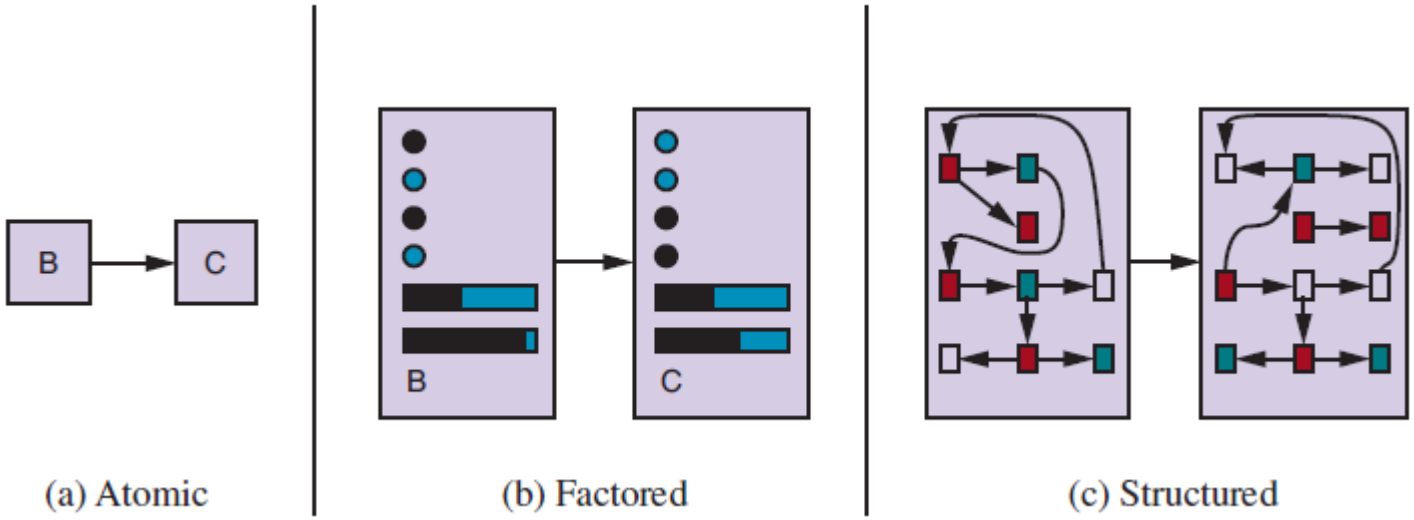
- A general learning agent - The "performance element" box represents what we have previously considered to be the whole agent program. Now, the "learning element" box gets to modify that program to improve its performance.



- The most important distinction is between the learning element, which is responsible for making improvements, and the performance element, which is responsible for selecting external actions.
- The performance element is what we have previously considered to be the entire agent: it takes in percepts and decides on actions.
- The learning element uses feedback from the critic on how the agent is doing and determines how the performance Critic element should be modified to do better in the future.
- The last component of the learning agent is the problem generator. It is responsible for suggesting actions that will lead to new and informative experiences.
- The performance standard distinguishes part of the incoming percept as a reward (or penalty) that provides direct feedback on the quality of the agent's behavior.

How the components of agent programs work

Representations along an axis of increasing complexity and expressive power - atomic, factored, and structured.



(a) Atomic (b) Factored (c) Structured

(a) Atomic representation: a state (such as B or C) is a black box with no internal structure;

(b) Factored representation: a state consists of a vector of attribute values; values can be Boolean, real valued, or one of a fixed set of symbols.

(c) Structured representation: a state includes objects, each of which may have attributes of its own as well as relationships to other objects.

- The axis along which atomic, factored, and structured representations lie is the axis of increasing expressiveness.
- Distributed representations are more robust against noise and information loss.